



Brief paper

Distributed stabilization control of rigid formations with prescribed orientation[☆]Zhiyong Sun^b, Myoung-Chul Park^c, Brian D.O. Anderson^{a,b}, Hyo-Sung Ahn^c^a School of Automation, Hangzhou Dianzi University, Hangzhou 310018, China^b Data61 - CSIRO and Research School of Engineering, Australian National University, Canberra ACT 2601, Australia^c School of Mechanical Engineering, Gwangju Institute of Science and Technology, Gwangju, Republic of Korea

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ABSTRACT

Most rigid formation controllers reported in the literature aim to only stabilize a rigid formation shape, while the formation orientation is not controlled. This paper studies the problem of controlling rigid formations with prescribed orientations in both 2-D and 3-D spaces. The proposed controllers involve the commonly-used gradient descent control for shape stabilization, and an additional term to control the directions of certain relative position vectors associated with certain chosen agents. In this control framework, we show the minimal number of agents which should have knowledge of a global coordinate system (2 agents for a 2-D rigid formation and 3 agents for a 3-D rigid formation), while all other agents do not require any global coordinate knowledge or any coordinate frame alignment to implement the proposed control. The exponential convergence to the desired rigid shape and formation orientation is also proved.

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1. Introduction

1.1. Background and motivation

Formation control for a group of autonomous mobile agents has gained much attention due to its broad applications in many areas including both civil and military fields. A key problem in this domain that receives particular interest is how to stabilize and maintain a geometrical formation shape in a distributed manner. In the recent survey paper (Oh, Park, & Ahn, 2015), different types of formation control strategies are reviewed and compared, among which two most commonly-used approaches are

- the linear displacement-based approach: the desired formation is specified by a certain set of inter-agent *relative positions*

which means that the orientation of the final formation is implicitly fixed;

- the nonlinear distance-based approach: the desired formation is specified by a certain set of inter-agent *distances*, and the orientation of the target formation is *not* implicitly or explicitly defined.

For the first approach, all the agents must have their coordinate bases with the same orientation (while the origins may be different) such that the desired relative position vectors are well defined and controlled between agents (see e.g. Ren & Beard, 2008; Xiao, Wang, Chen, & Gao, 2009). This means that *all* the agents should be equipped with a compass or equivalent to guarantee their coordinate orientation alignments, which may not be practical in real world applications since having consistency among the compass readings would be a challenging task due to instrument bias and/or local variations caused by, e.g., the earth's magnetic field. The coordinate frame requirement was largely ignored in early works on formation control (as reviewed in Oh et al., 2015). It is only in recent years that the importance of coordinate frame issue has been recognized in formation controller design and implementation. In the case that initially all the agents in the 2-D plane have different local coordinate frames, one needs to design a combined control establishing coordinate frame direction alignment and linear displacement-based formation stabilization to ensure the convergence to a target shape (Oh & Ahn, 2014b). Furthermore,

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it has also been shown in Meng, Anderson, and Hirche (2016) that the assumption that all the agents have coordinate systems with the same orientation may not be realistic in practice as small perturbations in their local coordinate systems will cause unexpected behaviors for the displacement-based formation system. Thus in practice, a coordinate-free formation control system is almost always preferable. In Aranda, López-Nicolás, Sagüés, and Zavlanos (2015), a coordinate-free formation control strategy was proposed by including a rotation matrix in the formation controller. The advantage of the coordinate-free property of the proposed formation controller in Aranda et al. (2015) is offset by the requirement that the relative position measurements from *all* other agents should be available to each individual agent, which implies that the coordinate-free formation control in Aranda et al. (2015) is effectively not a distributed one. Recent efforts also show that the bearing-based approach is another promising strategy to achieve a desired formation (Zhao & Zelazo, 2016). We note that such an approach however still does not resolve the strict requirement of the global knowledge of coordinate frame orientation among all agents.

All these disadvantages on the coordinate frame requirement can be avoided in the distance-based formation setup. This is because in the distance-based setup a global coordinate system defining a common orientation for all individual agents' coordinate frames is not required, and each agent can use its local coordinate basis to achieve a rigid formation shape (we refer the readers to Fig. 3 in Oh et al., 2015 for a comparison on the coordinate basis requirement for these two approaches). Rigid formation control has been discussed extensively in the literature, most of which has focused on the convergence analysis of formation shapes (see e.g. Anderson & Helmke, 2014; Cai & De Queiroz, 2015; Dorfner & Francis, 2010; Krick, Broucke, & Francis, 2009). Note however that in many applications involving multi-agent coordination, a formation with both a desired shape and a particular orientation is required. At the same time, for distance-based rigid formation control, the orientation of the final formation is not controlled and actually not well defined,¹ which may limit the practical application of shape controllers discussed in these previous works. In this paper, we aim to design distributed formation controllers to achieve a desired rigid formation with a prescribed formation orientation.

1.2. Related work on formation control with desired orientation

In the above subsection we have reviewed different approaches in formation control, and have emphasized the importance of local coordinate frames in implementing a formation controller. In this subsection we will further review some more related work on formation control with desired orientation. The stabilization control of rigid formations with desired orientation was discussed in Pais, Cao, and Leonard (2009) by using tensegrity theory and a projected collinear structure. However, the approach still requires all the agents to have knowledge of the orientation of a common reference frame. The problem of stabilizing only the orientation of rigid objects subject to distance constraints was

studied in Markdahl, Karayiannidis, Hu, and Kragic (2012) and Wang, Markdahl, and Hu (2011), by assuming that the rigid shapes have been achieved before applying the orientation control. Thus, the approaches in Markdahl et al. (2012) and Wang et al. (2011) cannot be applied to solve the formation stabilization control task in question. In our previous paper (Sun, Mou, Anderson, & Morse, 2014) we showed a feasible approach to move or re-orient a rigid formation to a desired orientation by introducing distance mismatches; however, such orientation control approach, which is a by-product of the mismatched formation control problem, indicates that the final formation is slightly distorted compared to the desired formation. Furthermore, the orientation control in Sun et al. (2014) also requires global information in terms of all other agents' positions which is contrary to the aim of using a distributed approach for the formation control task.

We note that a very general control framework for stabilizing an *affine* formation was recently proposed in Lin, Wang, Chen, Fu, and Han (2016), in which a strict assumption that the target formation should be *globally rigid* was imposed to generate a rigid shape with orientation constraint. Such an assumption is not required in the control strategy proposed in this paper. Also note that the formation orientation problem discussed in this paper is a stabilization control problem (i.e. the aim is to achieve a *static* formation with desired orientation), while a motion generation problem involving rigid formation orientation was discussed in Garcia de Marina, Jayawardhana, and Cao (2016) with a totally different control architecture.

1.3. Main contributions

In this paper we propose feasible and distributed controllers to achieve both rigid shape stabilization and formation orientation control with *minimal knowledge of global coordinate orientation* for the agent group. The basic idea underlying the controller design is to choose certain agents as *orientation agents* (definitions will become clear in the context), for which some of the associated relative position vectors should achieve both desired distances and directions specified in the global coordinate frame. Some preliminary results were presented in Park and Ahn (2015) and Sun and Anderson (2015). This paper extends the results reported in Park and Ahn (2015) and Sun and Anderson (2015), by providing a general and systematic approach to solve this control problem with a minimal number of orientation agents. Compared to Park and Ahn (2015) and Sun and Anderson (2015), the main extensions and contributions in this paper can be summarized as follows. First, the results to be discussed in this paper can be applied to stabilize rigid shapes and orientations without any restriction on agent numbers and ambient space dimensions, while Park and Ahn (2015) presented preliminary results focusing on proving asymptotic stability for a 2-D four-agent formation system. Second, we have removed the assumption that the target formation shapes are minimally rigid, which was a key assumption in Park and Ahn (2015) and Sun and Anderson (2015). Instead, by developing different approaches in the proofs, this paper only assumes that target formation shapes are infinitesimally rigid. Furthermore, by exploring several novel observations concerning the rigidity matrix from graph rigidity theory, we will also prove an *exponential* convergence to the desired formation shape with specified orientations. Note that the exponential stability renders the robustness property of the proposed formation control system in the presence of small measurement errors or perturbations.²

¹ We need to distinguish different meanings of *orientation* in the context of formation control. By regarding a rigid formation as a rigid body, the formation orientation relates to the overall rigid formation. The orientation concept in e.g. Montijano, Zhou, Schwager, and Sagues (2014) and Oh and Ahn (2014b) refers to the orientation of the *local coordinate frame* for each agent. We will distinguish different meanings by referring explicitly to either formation orientation or coordinate orientation. Another orientation concept refers to the definition of signed area for a closed curve formed by a formation shape with a specific ordering of all agents (e.g. a triangle with positive/negative area). This concept will not be used in this paper.

² The problem of securing robustness given massive measurement errors induced by sensor failure is different, and is discussed in Yu and Anderson (2009).

In summary, the main contributions are listed as follows:

- We propose feasible and distributed controllers to stabilize infinitesimally rigid target formations with desired orientations;
- The minimal knowledge of global coordinate frame is identified, for both 2-D and 3-D formations;
- We prove that the proposed controllers guarantee the exponential convergence to the target formation with a desired orientation.

2. Preliminaries and problem setup

2.1. Preliminary on graph theory

The notations used in this paper are fairly standard, and we refer the readers to [Sun, Park, Anderson, and Ahn \(2016\)](#) for a more detailed introduction on various notations used in this paper. Since formations of n mobile agents are best described in terms of graph theory, we give a brief description of some of the basic definitions and facts needed. Consider an undirected graph with m edges and n vertices, denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with vertex set $\mathcal{V} = \{1, 2, \dots, n\}$ and edge set $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$. The neighbor set $\mathcal{N}_i$ of node i is defined as $\mathcal{N}_i := \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$. The matrix relating the nodes to the edges is called the incidence matrix $H = \{h_{ki}\} \in \mathbb{R}^{m \times n}$, whose entries are defined as (with arbitrary edge orientations for undirected formations considered here): $h_{ki} = +1$ if the k th edge sinks at node i , or $h_{ki} = -1$ if the k th edge leaves node i , or $h_{ki} = 0$ otherwise. For a connected and undirected graph, one has $\text{rank}(H) = n - 1$ and $\text{null}(H) = \text{span}\{\mathbf{1}_n\}$.

2.2. Rigidity theory

Given a vertex element $i \in \mathcal{V}$ we associate to it a point p_i of Euclidean space $\mathbb{R}^d$.³ The column vector $p = [p_1^T, p_2^T, \dots, p_n^T]^T$ thus describes a *framework* $(\mathcal{G}, p)$ of n agents, labeled by the set of vertices⁴ of $\mathcal{G}$. For any edge $k \in \mathcal{E}$ with head j and tail i which is consistent with the construction of the matrix H , consider the associated relative position vector defined as $z_k = p_j - p_i$. Let

$$z = [z_1^T, z_2^T, \dots, z_m^T]^T \in \mathbb{R}^{dm}$$

$$Z(z) = \text{diag}(z_1, z_2, \dots, z_m) \in \mathbb{R}^{dm \times dm}$$

denote the associated column vector and block diagonal matrix, respectively. Note that there holds

$$z = (H \otimes I_d)p. \quad (1)$$

With this notation at hand, we consider the smooth distance map

$$r_{\mathcal{G}} : \mathbb{R}^{dn} \longrightarrow \mathbb{R}^m, \quad r_{\mathcal{G}}(p) = (\|p_i - p_j\|^2)_{(i,j) \in \mathcal{E}} = Z(z)^T z. \quad (2)$$

The rigidity of frameworks is then defined as follows.

Definition 1 ([Asimow & Roth, 1979](#)). A framework $(\mathcal{G}, p)$ is rigid in $\mathbb{R}^d$ if there exists a neighborhood $\mathcal{U}$ of p such that $r_{\mathcal{G}}^{-1}(r_{\mathcal{G}}(p)) \cap \mathcal{U} = r_{\mathcal{K}}^{-1}(r_{\mathcal{K}}(p)) \cap \mathcal{U}$ where $\mathcal{K}$ is the complete graph with the same vertex set as $\mathcal{G}$.

Two frameworks $(\mathcal{G}, p)$ and $(\mathcal{G}, \bar{p})$ are equivalent if $r_{\mathcal{G}}(p) = r_{\mathcal{G}}(\bar{p})$ and are congruent if $\|p_i - p_j\| = \|\bar{p}_i - \bar{p}_j\|$ for all $i, j \in \mathcal{V}$. A useful tool to study graph rigidity is the **rigidity matrix**, which is defined as the Jacobian matrix $R(p) = \frac{1}{2} \partial r_{\mathcal{G}}(p) / \partial(p)$. By inspection, $R(p)$ is an $m \times dn$ matrix given as

$$R(p) = Z(z)^T (H \otimes I_d). \quad (3)$$

Note that the entries of $R(p)$ only involve relative position vectors z , and we can rewrite it as $R(z)$. The rigidity matrix will be used to determine the infinitesimal rigidity of a framework, as shown in the following theorem.

Theorem 1 ([Hendrickson, 1992](#)). Consider a framework $(\mathcal{G}, p)$ in d -dimensional space with $n \geq d$ vertices and m edges. It is infinitesimally rigid if and only if

$$\text{rank}(R(p)) = dn - d(d+1)/2. \quad (4)$$

Specifically, the framework $(\mathcal{G}, p)$ is infinitesimally rigid in $\mathbb{R}^2$ (resp. $\mathbb{R}^3$) if and only if $\text{rank}(R(p)) = 2n - 3$ (resp. $\text{rank}(R(p)) = 3n - 6$). Obviously, in order to have an infinitesimally rigid framework, the graph should have at least $2n - 3$ (resp. $3n - 6$) edges in $\mathbb{R}^2$ (resp. $\mathbb{R}^3$).

From [Theorem 1](#), one knows that the dimension of the null space of $R(p)$ for an infinitesimally rigid framework $(\mathcal{G}, p)$ in the d -dimensional space is $d(d+1)/2$. The following lemma characterizes the structure of its null space.

Lemma 1 (*Null Space of the Rigidity Matrix*). Suppose the framework $(\mathcal{G}, p)$ is infinitesimally rigid with the associated rigidity matrix denoted as $R(p)$.

- The $d = 2$ case: The null space of $R(p)$ is of dimension 3 and is described as $\text{null}(R(p)) = \text{span}(q_1, q_2, q_3)$, where

$$q_1 = \mathbf{1}_n \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad q_2 = \mathbf{1}_n \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix};$$

$$q_3 = [(K_3 p_1)^T, (K_3 p_2)^T, \dots, (K_3 p_n)^T]^T,$$

and the matrix K_3 is defined as

$$K_3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

- The $d = 3$ case: The null space of $R(p)$ is of dimension 6 and is described as $\text{null}(R(p)) = \text{span}(q_1, q_2, q_3, q_4, q_5, q_6)$, where

$$q_1 = \mathbf{1}_n \otimes \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \quad q_2 = \mathbf{1}_n \otimes \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix};$$

$$q_3 = \mathbf{1}_n \otimes \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix};$$

$$q_i = [(K_i p_1)^T, (K_i p_2)^T, \dots, (K_i p_n)^T]^T, \quad i = 4, 5, 6;$$

and the matrix K_i is defined as

$$K_4 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}; \quad K_5 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix};$$

$$K_6 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Lemma 1 is a reformulation of some well known results on the rigidity matrix discussed in Tay and Whiteley's early paper (see [Tay & Whiteley, 1985](#), Page 31). We also note that the structure of the null space of a rigidity matrix for 3-D formations was re-stated in [Zelazo, Franchi, Bülthoff, and Robuffo Giordano \(2015, Theorem 2.16\)](#). The clear structure of the null space for a rigidity matrix

³ In this paper we will focus on 2-D and 3-D formations, i.e. $d = 2, 3$.

⁴ In this paper we assume each agent can be identified with a unique index, which is a realistic assumption in practice consistent with highlighting possible different roles of individual agents distinguished often by the sensing hardware carried. This will also facilitate the definition of a target formation with desired orientation, especially in the case of formation symmetry (e.g. a symmetric regular polygon formation).

in Lemma 1 also shows the relationship between null vectors of a rigidity matrix and invariant rigid motions (i.e. infinitesimal translations and rotations) for the associated rigid formation, and a clear proof of this point can be found in the online arXiv version (Sun et al., 2016). For instance, in the 3-D case, the null vectors q_1, q_2, q_3 correspond to the invariant translational motions, while the null vectors q_4, q_5, q_6 correspond to the invariant rotational motions. The structure of the null space will be particularly helpful in the controller design and useful in the stability analysis, as will be shown in the main part of this paper.

The infinitesimal rigidity also guarantees the following property for a framework.

Lemma 2. *Suppose the framework $(\mathcal{G}, p)$ is infinitesimally rigid. Then for any node i , the set of relative position vectors $p_j - p_i, j \in \mathcal{N}_i$ cannot all be collinear (in the 2-D case) or all be coplanar (in the 3-D case).*

The proof can be found in the full version of this paper (Sun et al., 2016). Lemma 2 will be useful for defining a feasible target formation by choosing some adjacent edges associated with certain agents (which will be discussed in Section 3.1).

2.3. Gradient-based formation controller and problem formulation

Let d_{kij} denote the desired distance of edge k which links agent i and j . We further define

$$e_{kij} = \|p_i - p_j\|^2 - d_{kij}^2 \quad (5)$$

to denote the squared distance error for edge k . In order to simplify the notation, we may use e_k and d_k interchangeably in the sequel. This will also apply to d_{kij} and d_k, z_k and z_{kij} in the following context when the dropping out of the dummy subscript ij in each vector causes no confusion. The squared distance error vector is denoted by $e = [e_1, e_2, \dots, e_m]^T$. In this paper, we suppose that each agent is modeled by a single integrator $\dot{p}_i = u_i$ where u_i is the controller to be designed for achieving the formation control objective.

In Krick et al. (2009), the following formation control system was proposed:

$$\dot{p}_i = - \sum_{j \in \mathcal{N}_i} (\|p_i - p_j\|^2 - d_{kij}^2)(p_i - p_j), \quad i = 1, \dots, n. \quad (6)$$

A highly non-trivial property for the above formation controller is its independence of a global coordinate frame, as summarized in the following lemma.

Lemma 3. *The controller for each agent as shown in (6) is independent of any global coordinates. That is, each agent can use its own coordinate system to measure relative positions and to implement the control.*

The proof can be found in e.g. Cao, Morse, Yu, Anderson, and Dasgupta (2011) and Krick et al. (2009) under different formation settings. Note that the control (6) is expressed in a global frame for notational simplicity, but the control action remains equivalent in any local coordinate frame. Also note that the above control describes a steepest descent gradient flow of the following potential function

$$V_1(p) = \frac{1}{4} \sum_{(i,j) \in \mathcal{E}} (\|p_i - p_j\|^2 - d_{kij}^2)^2. \quad (7)$$

This potential function (7) for rigid shape stabilization and the associated gradient flow (6) have been extensively studied in the literature (see e.g. Anderson & Helmke, 2014; Cao et al., 2011; Cortés, 2009; Dorfler & Francis, 2010; Krick et al., 2009; Oh & Ahn, 2014a). However, the above control and its extensions studied in

these previous papers only stabilize a rigid formation shape, while the orientation of the formation is not specified.

In this paper we consider the problem of how to simultaneously stabilize a rigid shape and achieve a desired orientation for a target formation. As will be seen in later analysis, the potential function (7) will play its role in stabilizing a rigid shape, but one needs to redefine a target formation by incorporating formation orientation information and then assign additional control terms to achieve this further formation control task.

3. Main result

3.1. Target formation and control framework

Before describing the controller design, we first discuss how to define a target formation with the given inter-agent distance and formation orientation constraints. As mentioned in the above section, the commonly-used gradient-based controller (6) does not control the orientation and there are certain degrees of freedom relating to rotations for a converged formation. Intuitively, by regarding the rigid formation as a rigid body and specifying certain directions of some chosen edges in a global coordinate frame, the orientation of the overall rigid formation can be fixed. This will be the basic idea in the definition of a target formation and the controller design discussed in the sequel.

For simplifying the controller design and implementation, we choose one agent and a certain number of its neighboring agents as the specified agents to implement the additional orientation control task, with the associated edges between them being assigned with both distance constraints and orientation constraints. We term these agents with the additional orientation control task as orientation agents, and other agents as non-orientation agents. Thus, the target formation is defined with inter-agent distance constraints applying to all the agents, and orientation constraints for the chosen edges between orientation agents. Since the orientation agents possess more information on a target formation compared with non-orientation agents and are tasked with additional control terms to steer the formation to a desired orientation, they could be considered as leader agents in the general context of distributed control (Oh et al., 2015; Ren & Beard, 2008).

For convenience in later analysis, we denote $\mathcal{G}_o$ as the underlying graph of the orientation control to distinguish it with the underlying graph $\mathcal{G}$ of the formation shape control. If the edge (i, j) associated with agent i and j is chosen in the orientation control in $\mathcal{G}_o$, we denote it as $(i, j) \in \mathcal{E}_o$. The set of neighboring agents of orientation agent i chosen in the orientation control is defined as $\mathcal{N}_i^o := \{j \in \mathcal{V} : (i, j) \in \mathcal{E}_o\}$. Note that $\mathcal{G}_o$ includes all vertices as $\mathcal{G}$, but it only contains the edges $\mathcal{E}_o$. The desired direction for the relative position vector $p_j - p_i$ for edge $(i, j) \in \mathcal{E}_o$ is denoted by a given vector $\hat{p}_{ji} := \hat{p}_j - \hat{p}_i$, where $\hat{p}_j, \hat{p}_i \in \mathbb{R}^d$ are orientation agents' target positions (which are translation invariant) that realize the specified relative position vector $\hat{p}_{ji}$. The relative position constraint should also be consistent with the distance constraint associated with the chosen edge (see Definition 2). Thus, the orientation control is used to additionally stabilize the relative position $p_j - p_i$ to the desired one $\hat{p}_j - \hat{p}_i$ with $(i, j) \in \mathcal{E}_o$. Due to the rigid body property of a desired rigid formation, the formation orientation can be determined by the directions of a certain set of desired relative position vectors. We show two examples, a 2-D four-agent rectangular formation and a 3-D tetrahedral formation depicted in Fig. 1, to illustrate the formation control framework.

Note that any two agents associated with one edge can be chosen as orientation agents, and there is no need to design a centralized algorithm for the selection of the orientation agents. To define a target formation with prescribed orientation, one can

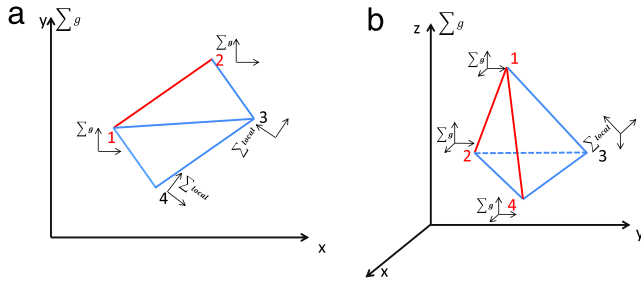


Fig. 1. Example of controlling a 2-D rectangular formation and a 3-D tetrahedral formation with prescribed orientation. (a) Agent 1 and one of its neighbors, agent 2, are chosen as orientation agents. The relative position vector $p_2 - p_1$ associated with edge (1,2) is used to describe the desired orientation, which is denoted by red color (in this example $(1, 2) \in \mathcal{E}_o$). (b) Agent 1 and two of its neighbors, agents 2 and 4, are chosen as orientation agents. The relative position vectors $p_2 - p_1$ and $p_4 - p_1$ associated with edges (1,2) and (1,4) are used to describe the desired orientation, which are denoted by red color (in this example $(1, 2), (1, 4) \in \mathcal{E}_o$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

first choose one agent and then select any one relative position vector (for 2-D formations), or any two non-collinear relative position vectors (for 3-D formations), from its adjacent edges to specify the desired formation orientation. According to Lemma 2, the existence of non-collinear or non-coplanar adjacent edges guarantees one can select any agent to define a target formation. To sum up, we give a formal definition of a target formation.

Definition 2 (Target Formation). The target formation is defined as $(\mathcal{G}, \tilde{p})$ which satisfies the following constraints

- Distance constraints: $\|\tilde{p}_i - \tilde{p}_j\| = d_{k_{ij}}, \forall (i, j) \in \mathcal{E}$;
- Orientation constraints: $\tilde{p}_i - \tilde{p}_j = \hat{p}_i - \hat{p}_j, \forall (i, j) \in \mathcal{E}_o$;

Note that there should hold $\|\hat{p}_i - \hat{p}_j\| = d_{k_{ij}}$ so that the orientation constraint is consistent with the formation shape constraint.

In order to well define the orientation constraint, we need the following alignment assumption.

Assumption 1. All orientation agents should be equipped with coordinate systems with the same coordinate axis directions parallel with those of the global coordinate system.

Take the formation control formulation in Fig. 1(a) as an example. Since agents 1 and 2 are chosen as orientation agents, their coordinate systems should be aligned with the global coordinate system denoted by Σ_g . Such a global coordinate system is required to define the desired relative position vector $(\hat{p}_j - \hat{p}_i)$ for $(i, j) \in \mathcal{E}_o$. Thus Assumption 1 provides a necessary condition for the controller design and implementation.⁵

3.2. Controller design

We propose the following formation stabilization controller:

$$\begin{aligned} \dot{p}_i(t) = & \underbrace{\sum_{j \in \mathcal{N}_i} (p_j(t) - p_i(t)) (\|p_j(t) - p_i(t)\|^2 - d_{k_{ij}}^2)}_{\text{shape control term, if } (i,j) \in \mathcal{E}} \\ & + \underbrace{\sum_{j \in \mathcal{N}_i^o} ((p_j(t) - p_i(t)) - (\hat{p}_j - \hat{p}_i))}_{\text{orientation control term, if } (i,j) \in \mathcal{E}_o}. \end{aligned} \quad (8)$$

⁵ We comment that, as shown with more detail in Sun et al. (2016), specification of a single orientation edge does not deal with the problem of reflection ambiguity in determining the formation using a set of prescribed distances. Since in this paper we only focus on local convergence, the reflection ambiguity will not be further discussed in the paper. We refer the readers to Sun et al. (2016) for more discussions on the formation reflection issue.

It is obvious from Eq. (8) that the proposed control is distributed since only local information from neighboring agents in terms of relative positions is needed. In the later analysis we will also show that the overall system consisting of n agents described by (8) is a gradient system associated with a cost function.

The above formation control system (8) can be written in a compact form

$$\dot{p} = -R^T e - (L_o \otimes I_d) \tilde{p} \quad (9)$$

where L_o is the Laplacian matrix of the underlying undirected graph $\mathcal{G}_o$ for the orientation control, and the vector $\tilde{p} = [\tilde{p}_1^T, \tilde{p}_2^T, \dots, \tilde{p}_n^T]^T$ is defined as $\tilde{p}_i = p_i - \hat{p}_i$ if i is a chosen orientation agent,⁶ or $\tilde{p}_i = \mathbf{0}$ otherwise.

For the formation control system (9), the set of the desired equilibrium is described as

$$\mathcal{M} = \{p \in \mathbb{R}^{dn} | e(p) = \mathbf{0}, p_i - p_j = \hat{p}_i - \hat{p}_j, \forall (i, j) \in \mathcal{E}_o\} \quad (10)$$

which satisfies the constraints in Definition 2.

3.3. Properties of the formation control system

In the following, we show several properties of the proposed control (8).

Lemma 4. The position of the formation centroid is preserved by the above control law (8).

The proof can be found in Sun et al. (2016). Lemma 4 corresponds to the statement of invariant centroid in rigid formation control without orientation control, as proved in e.g. Krack et al. (2009).

In the spirit of Lemma 3, we claim the second property of the proposed controllers.

Lemma 5. For all non-orientation agents, their local coordinate systems are sufficient to implement the control law.

The above property for the proposed controller is a direct consequence of Lemma 3 on gradient descent control of rigid formations. This controller property has been illustrated in Fig. 1. In the example shown in Fig. 1(a), agent 3 or 4 is not an orientation agent and its coordinate system orientation does not need to be aligned with the global coordinate system. As a consequence of Lemma 5, the minimum number of orientation agents is 2 for a 2-D rigid formation and 3 for a 3-D rigid formation. Hence the minimum number of agents requiring knowledge of the global coordinate frame for the multi-agent formation group is 2 or 3, according as the 2-D or 3-D case being considered.

3.4. Convergence analysis

We will first show that the gradient property of the proposed controller and a general result on the convergence.

Theorem 2. The formation control system with the proposed controller (8) describes a gradient control system and the formation system converges to the largest invariant set in the set $\mathcal{O}(z)$ defined as

$$\mathcal{O}(z) = \{z | R(z)^T e(z) + (L_o \otimes I_d) \tilde{p}(z) = 0\}. \quad (11)$$

⁶ Note that the vector $\tilde{p}_i$ is not an actual control input as $\hat{p}_i$ may not be available for agent i (the actual control term is $\tilde{p}_{ji} := \hat{p}_j - \hat{p}_i$). The introduction of $\tilde{p}_i$ is for the convenience of analysis and for writing a compact form of the formation system as in (9).

Proof. We choose the same potential function in (7) as the potential for the shape control, and the following potential function

$$V_2 = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}_0} \|(p_j(t) - p_i(t)) - (\hat{p}_j - \hat{p}_i)\|^2$$

$$= \frac{1}{2} \bar{p}^T (L_o \otimes I_d) \bar{p} \quad (12)$$

for the orientation control. The composite potential function is then defined as $V(p) = V_1(p) + V_2(p)$. Note also that both V_1 and V_2 can be regarded as functions of z , and we will also write them for later analysis as $V_1(z)$ and $V_2(z)$, as well as $V(z)$.

The dynamical system for the relative position z defined in (1) is

$$\dot{z} = (H \otimes I_d) \dot{p}$$

$$= -(H \otimes I_d) R^T e - (H \otimes I_d) (L_o \otimes I_d) \bar{p} \quad (13)$$

and the distance error system is described by

$$\dot{e} = 2R\dot{p} = -2RR^T e - 2R(L_o \otimes I_d) \bar{p}. \quad (14)$$

Note that (13) constitutes a self-contained system since both e and $\bar{p}$ are expressible as functions of z . Note further that the potential functions V_1 and V_2 are functions involving only relative position vectors in terms of z and e rather than the absolute position vector p .⁷ Thus, we can write the potential as $V(z)$ for the self-contained z system (13). We then calculate the derivative of the potential V_1 and V_2 along the trajectories of system (13) and (14):

$$\dot{V}_1 = \frac{1}{2} e^T \dot{e} = e^T (-RR^T e - R(L_o \otimes I_d) \bar{p})$$

$$= -e^T RR^T e - e^T R(L_o \otimes I_d) \bar{p} \quad (15)$$

and

$$\dot{V}_2 = \bar{p}^T (L_o \otimes I_d) \dot{\bar{p}}$$

$$= \bar{p}^T (L_o \otimes I_d) (-R^T e - (L_o \otimes I_d) \bar{p})$$

$$= -\bar{p}^T (L_o \otimes I_d) R^T e - \bar{p}^T (L_o \otimes I_d) (L_o \otimes I_d) \bar{p} \quad (16)$$

where in the second equality we have used the non-trivial result $(L_o \otimes I_d) \dot{\bar{p}} = (L_o \otimes I_d) \bar{p}$. The derivative of V can be computed as

$$\dot{V} = \dot{V}_1 + \dot{V}_2$$

$$= -e^T RR^T e - e^T R(L_o \otimes I_d) \bar{p}$$

$$- \bar{p}^T (L_o \otimes I_d) R^T e - \bar{p}^T (L_o \otimes I_d) (L_o \otimes I_d) \bar{p}$$

$$= -e^T RR^T e - 2e^T R(L_o \otimes I_d) \bar{p}$$

$$- \bar{p}^T (L_o \otimes I_d) (L_o \otimes I_d) \bar{p}$$

$$= - (R^T e + (L_o \otimes I_d) \bar{p})^T (R^T e + (L_o \otimes I_d) \bar{p})$$

$$\leq 0. \quad (17)$$

The above derivative calculation thus implies that the formation system (8) describes a gradient descent flow for the composite potential V . Furthermore, the sub-level set of the potential $V(z)$ is compact with respect to the self-contained z system (13). By the LaSalle Invariance Principle, the solution of the formation system (13) converges to the largest invariant set in the set $\mathcal{O}(z) = \{z | \dot{V} = 0\}$ described in (11). $\square$

Theorem 2 focuses on the convergence of the z system (13) and its invariant set property, while the convergence of the z system (13) to the invariant set (11) does not directly guarantee the convergence of the position system (9) to a point. We remark that this stronger convergence result (the convergence of the position system to a point) can be shown by using the convergence property of the gradient system (see Absil & Kurdyka, 2006) and the real analyticity of the potential $V(p)$, considered as a function of p . A detailed discussion on the property of the equilibrium set $\mathcal{O}(z)$ is provided in Sun et al. (2016).

Similar to most works on rigid formation stabilization, in the following we will focus on local convergence analysis. In particular, we aim to show that the convergence to the target formation with desired distances and orientation is *exponentially fast*. The analysis is based on the linearization technique. We first compute the Jacobian of the vector field on the right-hand side of (9) around a desired equilibrium $\bar{p} \in \mathcal{M}$:

$$J_f = \left. \frac{\partial (-R^T e - (L_o \otimes I_d) \bar{p})}{\partial p} \right|_{p=\bar{p}}$$

$$= - \left. \frac{\partial R^T e}{\partial p} \right|_{p=\bar{p}} - R^T \left. \frac{\partial e}{\partial p} \right|_{p=\bar{p}} - \left. \frac{\partial (L_o \otimes I_d) \bar{p}}{\partial p} \right|_{p=\bar{p}}$$

$$= -(R^T R + (L_o \otimes I_d))|_{p=\bar{p}} \quad (18)$$

where $\left. \frac{\partial R^T e}{\partial p} \right|_{p=\bar{p}} = \mathbf{0}$ due to the fact that $e(p) = \mathbf{0}$ for a point p in the equilibrium set $\mathcal{M}$, and $\left. \frac{\partial e}{\partial p} \right|_{p=\bar{p}} = R$ according to the definition of the rigidity matrix.

Thus, the linearization equation of (9) is described as

$$\delta \dot{p} = -(R^T R + L_o \otimes I_d) \delta p. \quad (19)$$

In the following, we prove that the convergence is exponentially fast.

Theorem 3. Suppose the target formation is infinitesimally rigid and initial positions of all the agents are chosen such that the initial formation is close to the desired formation. With the proposed control law (8), the convergence to the correct formation shape and orientation is exponentially fast.

Before giving the proof of the above result, we first show a key lemma on the property of the linearization matrix.

Lemma 6. Suppose the target formation is infinitesimally rigid and the orientation edges are selected according to Section 3.1. Then the linearization matrix $\mathcal{F} := R^T R + L_o \otimes I_d$ is positive semidefinite and has d zero eigenvalues. Furthermore, there holds $\text{null}(\mathcal{F}) = \text{null}(H \otimes I_d) = \text{span}(\mathbf{1}_n \otimes I_d)$.

The proof can be found in the full version of this paper (Sun et al., 2016).

Proof of Theorem 3. As shown in Lemma 4, the formation centroid is stationary. We construct an orthogonal matrix $Q \in \mathbb{R}^{dn \times dn}$ whose first d rows are $\frac{1}{\sqrt{n}}(\mathbf{1}_n \otimes I_d)^T$. With Q , one can perform the coordinate transform on p as

$$\tilde{p} = Qp = \begin{bmatrix} p^o \\ p_r \end{bmatrix} \quad (20)$$

where $p^o = \sqrt{n}p_c$ and p_c is the formation centroid position (i.e. $p_c = \frac{1}{n} \sum_{i=1}^n p_i = \frac{1}{n}(\mathbf{1}_n \otimes I_d)^T p$). From Lemma 4, one has $\dot{p}^o = \sqrt{n}\dot{p}_c = \mathbf{0}$. We also define a reduced transformation matrix $Q_r \in \mathbb{R}^{d(n-1) \times dn}$, obtained from Q by removing the first d rows. Note that there holds $p = Q^{-1}\tilde{p} = Q^T \tilde{p}$ and $p_r = Q_r p$. For the

⁷ Also note that the distance error vector e can be written in terms of z according to the definition of e in (5).

linearized system (19), one can obtain the following coordinate-transformed system

$$\begin{aligned} \begin{bmatrix} \delta \dot{p}^o \\ \delta \dot{p}_r \end{bmatrix} &= Q \dot{\delta p} = -Q(R^T R + L_o \otimes I_d) \delta p \\ &:= -Q \mathcal{F} Q^{-1} \delta \tilde{p}. \end{aligned} \quad (21)$$

According to the structure of the matrix Q , there holds

$$\begin{aligned} Q \mathcal{F} Q^{-1} &= \begin{bmatrix} \frac{1}{\sqrt{n}} (\mathbf{1}_n \otimes I_d)^T \\ Q_r \end{bmatrix} \mathcal{F} \begin{bmatrix} \frac{1}{\sqrt{n}} (\mathbf{1}_n \otimes I_d) Q_r^T \\ \mathbf{0} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & Q_r \mathcal{F} Q_r^T \end{bmatrix}. \end{aligned} \quad (22)$$

Therefore,

$$\begin{aligned} \delta \dot{p}^o &= \mathbf{0} \\ \delta \dot{p}_r &= -Q_r \mathcal{F} Q_r^T \delta p_r. \end{aligned} \quad (23)$$

According to the definition of Q_r , the range space of Q_r^T is the orthogonal complement of the subspace $\text{span}(\mathbf{1}_n \otimes I_d)$. This, together with Lemma 6, implies that the linearization matrix $-Q_r \mathcal{F} Q_r^T$ is negative definite. Thus the convergence to the origin for the system (22) is locally exponentially fast. Since the system (22) is obtained from the system (19) by a linear coordinate transformation described in (20), the above statement also implies that the convergence to a point in the desired equilibrium $\mathcal{M}$ for the original system (9) is locally exponentially fast (Khalil, 1996, Theorem 4.13). For the linearized system, the guaranteed exponential convergence rate obtained in the linearization analysis is $\gamma = \lambda_{\min}(Q_r \mathcal{F} Q_r^T)$. Note that by the Courant–Fischer Theorem (Zhang, 2011, Theorem 8.9), the rate $\gamma = \lambda_{\min}(Q_r \mathcal{F} Q_r^T)$ is the same as the smallest positive eigenvalue of $\mathcal{F}$. $\square$

We refer the readers to Sun et al. (2016) for several discussions and remarks on the assumption of infinitesimal rigidity and consequences of exponential stability in the main result proved above. Also, several illustrative examples and simulations are provided in Sun et al. (2016) which support the above theoretical results.

4. Conclusion

In this paper we have discussed the formation control problem to achieve both a desired rigid shape and a formation orientation in ambient two-dimensional and three-dimensional spaces. The designed controller combines the advantages of the displacement-based approach and the distance-based approach, by specifying a small number of agents as orientation agents which are tasked to control associated relative position vectors to desired directions. The proposed controllers are distributed in that only relative position measurements from neighboring agents are required. For all non-orientation agents, no information about the global coordinate system is required for them to implement their respective controls, which guarantees a minimal requirement of the global knowledge of the global coordinate system.

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